

Nothing is landing on main

Everyone branched correctly. That was never the problem. The problem is that **25 branches have been growing for five weeks and only one has ever merged**. Here is what collides, and the order that gets us through it.

BRANCHES OPEN	MERGED INTO MAIN	COMMITTS WRITTEN	WRITTEN THIS WEEK
25	1	158	82%
One per person or feature	PR #37. Four other PRs merged sideways into feature branches.	Across all branches	130 of 158 commits

01 – THE SHAPE OF IT

Individually fine. Together, a pile-up.

Every branch was trial-merged against main, then replayed one at a time to see what happens as work actually lands. Nothing below touched the repo.

Tested on its own against main, almost everything is clean — **20 of 23 branches merge without a conflict**. That is the number people check, and it is why this looks fine from the inside.

Replay them in sequence, though, and only **8 stay clean**. The other 13 break, because they are not conflicting with main — they are conflicting with *each other*, and that only shows up once someone else goes first.

Main itself is four commits and one merge. Everything else the team built this semester is still sitting outside it.

02 – FOUND WHILE CHECKING

Two things worth fixing before anyone merges

Neither of these is a conflict yet. Both will become one.

Two branches are byte-for-byte identical

```
git rev-list --left-right --count \  
    origin/12-create-basic-enemy...origin/Enemy-Behaviour  
0          0
```

`12-create-basic-enemy` and `Enemy-Behaviour` have the same commits and zero difference between them. Both are Zarif's. **Pick one, delete the other** — otherwise the enemy work gets reviewed twice and merged twice.

Four of the five merged PRs went sideways, not to main

```
git log --all --grep "Merge pull request"  
#37 minigames/scaffold      → main  
#52 Splendid-T-textBoxes    → feat-UI-sprint1  
#51 task/pause-menu         → feature/ui-story  
#48 task/pause-menu         → feature/ui-story, player-HUD  
#47 task/pause-menu         → feature/ui-story, player-HUD
```

The team *is* reviewing and merging PRs — but into each other's feature branches instead of into main. Luke's pause commit `04a8fc3` now lives on `task/pause-menu`, `feature/ui-story` and `player-HUD`, having travelled there through PRs #47 and #48.

This is the actual root cause. Sideways merges make feature branches depend on each other, so no branch can land alone and main falls further behind with every PR. Merging into main would have shared that same work with all 25 branches at once.

03 — WHERE IT BREAKS

Three collision clusters

The 13 failures are not scattered. They land on three groups of files, and the people inside each group need to talk to each other before they open a PR.

GdxGame.java – screen registration

5 branches

level-1 s45855377

player-HUD Ayooluwa

task/pause-menu Luke

feature/ui-story Luke

minigames/Cyclops... 06wtkelly

Five branches register a new screen in the same enum, on the same lines. Three of them already conflict with main today, before anything else lands.

GdxGame.java

CONTENT

MainMenuActions.java

CONTENT

MinigameSelectActions.java

CONTENT

Enemy factory & configs

4 branches – worst cluster

Enemy-Behaviour Zarif

12-create-basic-enemy Zarif · dupe

enemy/enemy-health Jordan

ranged-attack Zarif

These are add/add conflicts — meaning several people created the *same file from scratch*, separately. Git has no shared history to merge from, so every one of these has to be resolved by hand. This cluster will hurt the most.

EnemyFactory.java

ADD/ADD

GhoulConfig.java

ADD/ADD

ProjectileFactory.java

ADD/ADD

NPCConfigs.java

CONTENT

assets/configs/NPCs.json

CONTENT

feature/player/movement DarcyMF

feature/player-combined dhirenmalik

feat/item&inventory wei weng

feature/player-item-interaction Aayush

player/player-ranged-attack dhirenmalik

feature/player-combined also hits add/add on `PlayerAttackComponent`, `ArrowProjectileComponent` and three test files — the same duplicate-creation problem as the enemy cluster.

PlayerFactory.java

CONTENT

PlayerActions.java

CONTENT

KeyboardPlayerInputComponent.java

CONTENT

PlayerAttackComponent.java

ADD/ADD

ArrowProjectileComponent.java

ADD/ADD

04 — THE ORDER

Merge in this sequence

Order matters here because each merge changes what the next one has to resolve. Going cluster by cluster keeps every conflict small enough for one person to fix.

STAGE 1 • TODAY

Land the eight that are already clean

These merge into main with no conflict, right now, and they shrink the surface for everything after:

`task/main-menu`, `feat-UI-sprint1`, `Splendid-T-textBoxes`, `JTrenerry-patch-1`, `enemy/delayed-attack`, `minigames/spin-the-wheel`, `player/player-ranged-attack`, and `feature/minigames` (already at main's tip).

One catch: `player/player-ranged-attack` and `feature/player-item-interaction` both touch `PlayerFactory.java`. Whichever goes second resolves it — **dhiren and Aayush, decide who goes first.**

STAGE 2 • TODAY, 2 MINUTES

Delete what is already dead

`task-winlose-screen` and `player-sprite-animations` have been untouched since 21 July and carry zero commits ahead of main. Also drop Zarif's duplicate enemy branch. Three fewer branches to reason about.

STAGE 3

GdxGame branches, strictly one at a time

After the first one merges, the other four each run `git merge origin/main` and fix their own enum conflict before opening a PR. It is a few lines each — but only if they do not all arrive at once.

STAGE 4 • NEEDS A CONVERSATION FIRST

Enemy cluster — agree on the files before merging

Zarif, Jordan, Ned: do not open these PRs cold. Decide together which version of `EnemyFactory.java` and `GhoulConfig.java` survives, have one person push the agreed version, and rebase the others onto it. Resolving add/add conflicts inside a PR review will go badly.

STAGE 5

Player cluster

`feature/player-item-interaction` first (it is clean against main), then `feature/player/movement`, `feat/item&inventory`, and `feature/player-combined` last — it carries the most add/add conflicts and benefits from everything else being settled.

Every branch, where it stands

Ahead and *behind* are relative to `origin/main`. *Cluster* is where the branch will collide.

BRANCH	OWNER	LAST COMMIT	AHEAD	BEHIND	CLUSTER
feat/item&inventory	wei weng	29 Aug	27	6	PLAYER
Enemy-Behaviour	ZarifHilman	30 Aug	25	6	ENEMY
12-create-basic-enemy	ZarifHilman	30 Aug	25	6	DUPLICATE
feature/ui-story	Luke Strader	27 Aug	25	1	GDXGAME
minigames/CyclopsMinigame/TimingBarComponents	O6wtKelly	29 Aug	24	1	GDXGAME
player-HUD	Ayooluwa Fadeyibi	27 Aug	24	6	GDXGAME
task/pause-menu	Luke Strader	27 Aug	21	6	GDXGAME
enemy/enemy-health	Jordan Conis	30 Aug	20	6	ENEMY
feature/player-item-interaction	AayushSinghParmar	26 Aug	20	6	PLAYER
level-1	s45855377	30 Aug	16	6	GDXGAME
feature/player-combined	dhirenmalik	29 Aug	16	0	PLAYER
feature/player/movement	DarcyMF	29 Aug	12	6	PLAYER
minigames/spin-the-wheel	Lachlan-Holliday	30 Aug	11	0	CLEAN
ranged-attack	ZarifHilman	29 Aug	9	6	ENEMY
enemy/delayed-attack	nedlonghurst	26 Aug	7	6	CLEAN
task/main-menu	soomayayaa	25 Aug	5	1	CLEAN
feat-UI-sprint1	Kinsey Dodd	27 Aug	2	1	CLEAN
Splendid-T-textBoxes	Kinsey Dodd	27 Aug	1	1	CLEAN
JTrenerry-patch-1	Jackson Trenerry	27 Aug	1	1	CLEAN
player/player-ranged-attack	dhirenmalik	26 Aug	1	1	CLEAN
feature/minigames	Jackson Trenerry	27 Aug	0	0	AT MAIN
task-winlose-screen	Jackson Trenerry	21 Jul	0	6	DELETE
player-sprite-animations	Jackson Trenerry	21 Jul	0	6	DELETE

One habit, starting now

Every time a PR merges into main, everyone runs `git merge origin/main` on their own branch that day.

That single habit is the whole fix. Most branches are only six commits behind — a two-minute merge today. The reason this became a pile-up is that nobody has synced since PR #37 landed, so 25 branches have been drifting in parallel with no idea what the others were touching.

Second habit: **point every PR at main**. Four of the five PRs merged so far went into another feature branch. It feels faster — you get your teammate's code immediately — but it welds two branches together, and neither can land on its own afterwards. A PR into main shares that same work with all 25 branches instead of just one.

Third habit: **open the PR small and early**. A 27-commit PR touching `PlayerFactory`, `ForestGameArea` and `EntityService` is not something a teammate can meaningfully review — so it sits, and drifts further.

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Method: every branch trial-merged with `git merge-tree --write-tree`, then replayed in sequence via `git commit-tree`. Simulation only — no branch, ref, or working tree was modified.